

JOY KUMAR GHOSH

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Profile Summary

I am Computer Science graduated with a substantial research focus on **quantum information** and **secure communication**, including **quantum key distribution** and **quantum circuit simulation**. Actively involved in research writing and computational experimentation, with **two manuscripts under review**. My **Research Interests**:

- i. **Quantum Cryptography for Secure Communication**
- ii. **Quantum Machine Learning**

Education

North South University (NSU) B.Sc. in Computer Science and Engineering Distinction: magna cum laude CGPA: 3.77/4.00 ($\approx 94\%$ marks)	Jan 2022 – Sep 2025 Dhaka, Bangladesh
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Birshreshtha Munshi Abdur Rouf Public College (BMAPRC) Higher Secondary Certificate (HSC) GPA: 4.33 out of 5.00 (Science)	Passing Year - 2019 Dhaka, Bangladesh
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Jadunath Pilot High School & College Secondary School Certificate (SSC) GPA: 5.00 out of 5.00 (Science)	Passing Year - 2017 Nagarpur, Tangail, Bangladesh
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Experience

Oracle APEX Web Application Developer – Intern MySoft Limited - Developed Outpatient Appointment System using APEX, and PL/SQL. [Project Link] - Participated in SRS Meetings with Client.	(3 Months) Aug 2025 – Nov 2025 Dhaka, Bangladesh
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Peer-Reviewed Journal Publications

1. **“Enhanced Selection Mechanism for Combined Quantum Key Distribution Protocol.”** 1st Author, under review in *APL Quantum*.
2. **“Accelerating Noisy Quantum Circuit Simulation via Unitary Precomputation.”** 1st Author, under review in *AIP Advances*.

Projects

1. **Undergraduate Capstone Project Showcase:** Enhanced Selection Mechanism for Entangled and Non-entangled Based Combined Quantum Key Distribution. Spring 2025. [\[Poster\]](#)
2. **Undergraduate CSE445 (Machine Learning) Course Project:** A thorough examination of Dhaka City’s weather data in Bangladesh, along with predictions of temperature, humidity, solar energy, solar radiation, and conditions through machine learning algorithms. Summer 2025. [\[Report\]](#)
3. **Undergraduate CSE465 (Pattern Recognition and Neural Network) Course Project:** Tomato Leaf Disease Detection using CNN and Lime as XAI. Spring 2025. [\[Report\]](#) [\[Repository\]](#)
4. **Undergraduate CSE425 (Concepts of Programming Language) Course Project:** Bangla Programming Language, Compiler Design with ANTLR. Summer 2024. [\[Report\]](#) [\[Repository\]](#)
5. **Undergraduate CSE311 (Database Systems) Course Project:** Nationwide Centralized Hospitals & Patients Management System, with MySQL, and PHP. Summer 2023. [\[Report\]](#) [\[Repository\]](#)

Selected Certifications

1. **Mahdy Research Academy:** Quantum Computing & Information. [Certificate]
2. **University of Colorado:** Applied Cryptography, Specialization. [[Certificate](#)]
3. **Duke University:** Explainable AI (XAI), Specialization [[Certificate](#)]
4. **KAIST:** Introduction to Quantum Information. [[Certificate](#)]

Skills

- **Quantum Tools:** Qiskit, TensorFlow Quantum.
- **Programming Language:** Python, C/C++, Java.
- **Machine Learning Tools:** PyTorch, Scikit-learn, Numpy, Pandas, LIME.
- **DevOps:** GitHub.
- **Writing Tools:** Latex and Office Suits.
- **Language:** English (Professional), Bangla (Native).

Honors & Awards

1. **Magna Cum Laude**, awarded by North South University. **Sep 2025**
2. **100% Merit-Based Scholarship** for Academic Excellence in Undergraduate. **Spring 2023 – Summer 2025**